

OV50E40 PDAF Application Note

V1.0.1

Safe Harbor Statement

- Except for historical information, the matters discussed in this presentation may contain forward-looking statements. These forward-looking statements are only estimates, and are subject to risks and uncertainties that may affect actual outcomes. Actual events or results may differ materially, and OmniVision disclaims any obligation to update information contained in any forward-looking statement.

Revision History

Revision	Date	Description
1.0.0	10.25.2022	Initial version Note: the register is just for checking exist setting to know PD output. If want to change to different PD output, please get setting from us.
1.0.1	01.10.2023	Update page 26

Glossary

- ✓ PD: Phase Detection
- ✓ PDAF: Phase Detection Auto Focus
- ✓ QPD: Quad PD. The whole array uses one Micro Lens covering 2x2 pixels.
- ✓ TL, TR, BL, BR: Top-Left, Top-Right, Bottom-Left, Bottom-Right
- ✓ L, R: Left, Right
- ✓ G1/2/3/4, R1/2/3/4, B1/2/3/4: 2x2 pixels index for top-left, top-right, bottom-left, bottom-right
- ✓ Windowing: output PD within a specific window
- ✓ VSkip2: in two vertical direction pairs keep one and drop another one
- ✓ VSkip4: in four vertical direction pairs keep one and drop other 3
- ✓ HSkip2: in two horizontal direction pairs keep one and drop another one
- ✓ VBin2: two vertical direction pairs bin together. It can be sum or average.

QPD Mode Summary

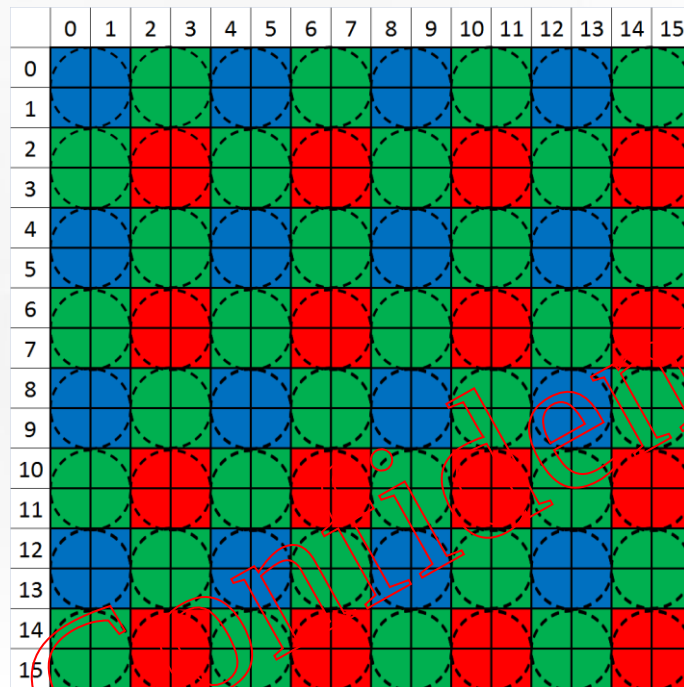
Mode	Analog	Sensor Output	Type 2 PD Output
Full / Cropping Bayer	Full / Cropping	Full / Cropping Bayer Raw (RM On)	4 PD (1/2/3/4)
Cropping Bayer Stagger HDR		Quarter Size Raw * 2	4 PD (1/2/3/4)
4C2+/4C2 Bin (2x2 Bin)	2x V Bin	Quarter Size Raw	2 PD (L/R)
4C2+/4C2 DCG (2x2 Bin)	2x V Bin	Combined Quarter Size Raw or Quarter Size Raw * 2	4C2+: 2 PD (L/R) from HCG 4C2: 2 PD (L/R) from HCG or LCG (programmable)
4C2+/4C2 DCG + VS (2x2 Bin)	2x V Bin	Combined Quarter Size Raw + VS Quarter Size Raw or Quarter Size Raw * 3	4C2+: 2 PD (L/R) from HCG or VS (programmable) 4C2: 2 PD (L/R) from HCG or LCG or VS (programmable)
4C2+/4C2 Stagger HDR (2x2 Bin)	2x V Bin	Quarter Size Raw * 2 or 3	2 PD (L/R) from one exposures (programmable)
4C2+/4C2 Sequential HDR (2x2 Bin)	2x V Bin	Quarter Size Raw * 2	2 PD (L/R) from both exposures or from one exposures (programmable)
4C1 Bin (2x2 Bin)	2x2 Bin	Quarter Size Raw	N/A
16C2+/16C2 Bin (4x4 Bin)	4x V Bin	1/16 Size Raw	2 PD (L/R)
16C1 Bin (4x4 Bin)	4x V Bin + 2x H Bin	1/16 Size Raw	N/A

PD Pattern and Alignment -- Full Size

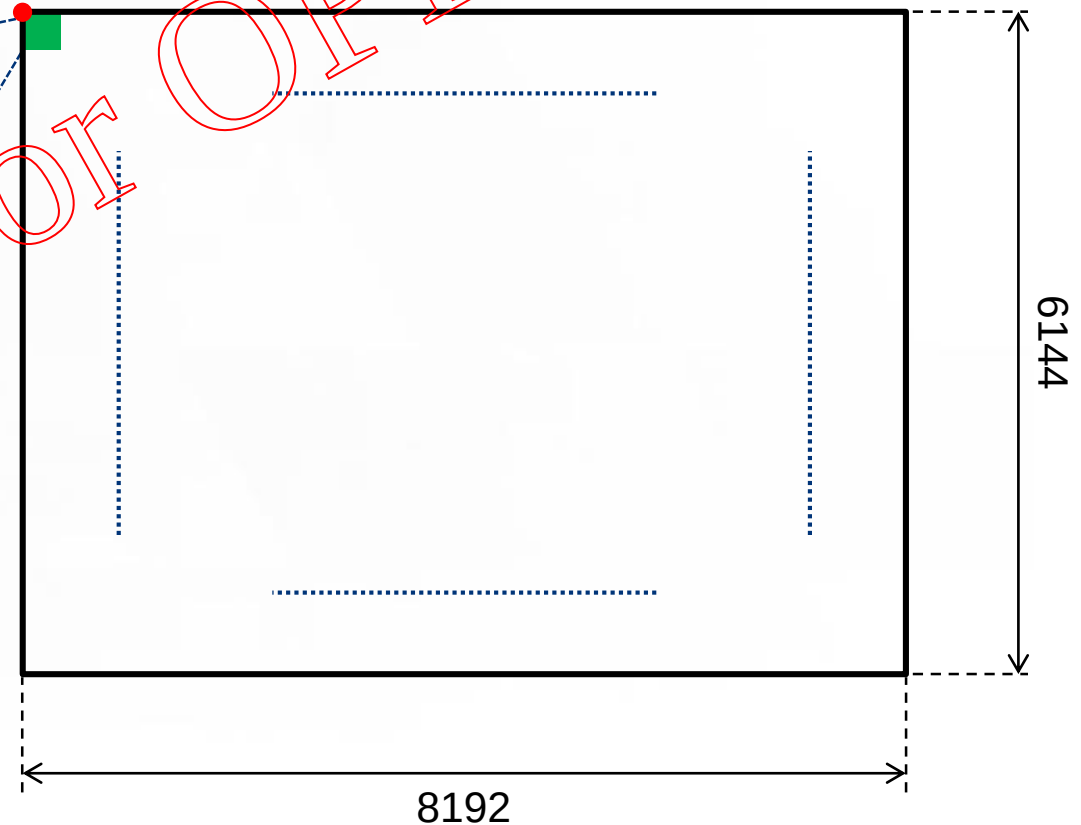
- ✓ Each 2x2 pixels are covered by one ML.
- ✓ Full size has four types of PD pixels, G1/2/3/4, R1/2/3/4, B1/2/3/4. Sensor has different options to output different amount of PD.

Normal

One block = multiple of 4x4 pixels
PD Density = 100%



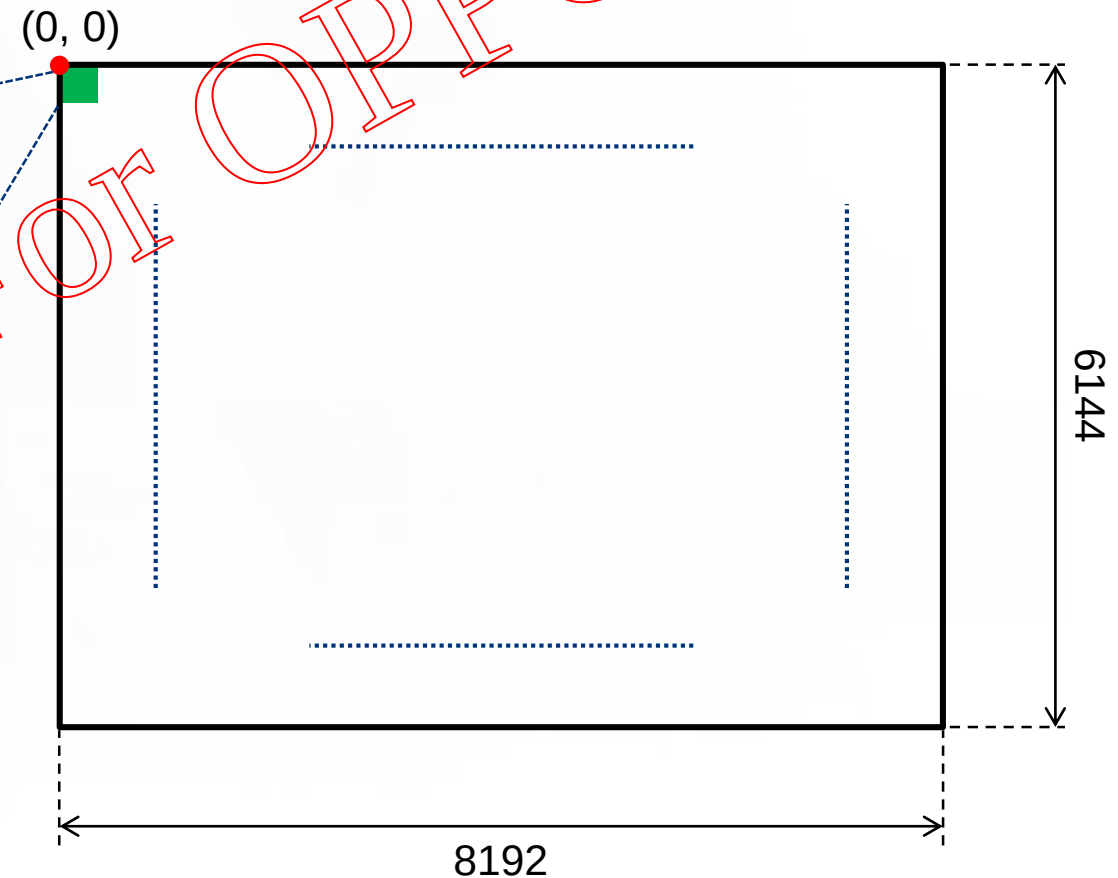
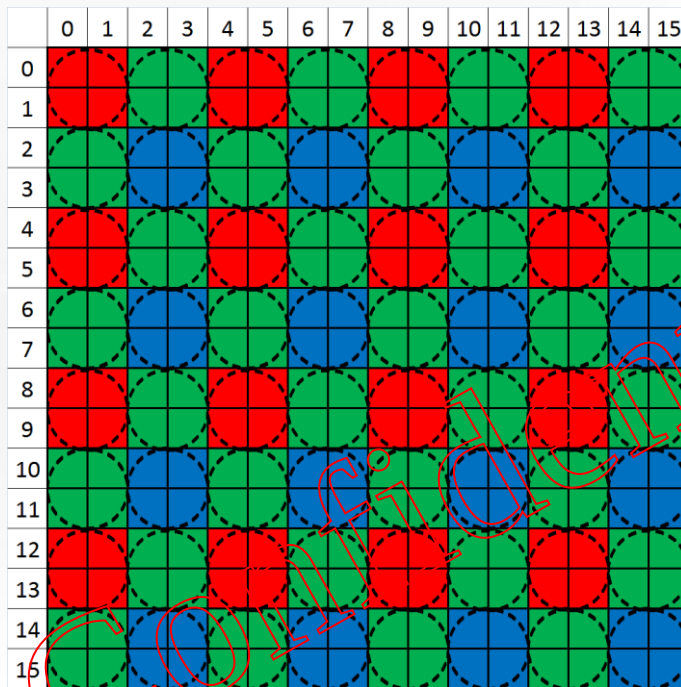
(0, 0)



PD Pattern and Alignment -- Full Size

- ✓ Each 2x2 pixels are covered by one ML.
- ✓ Full size has four types of PD pixels, G1/2/3/4, R1/2/3/4, B1/2/3/4. Sensor has different options to output different amount of PD.

One block = multiple of 4x4 pixels
Mirror&Flip PD Density = 100%



PD Pattern and Alignment -- 4C2 Binning Size

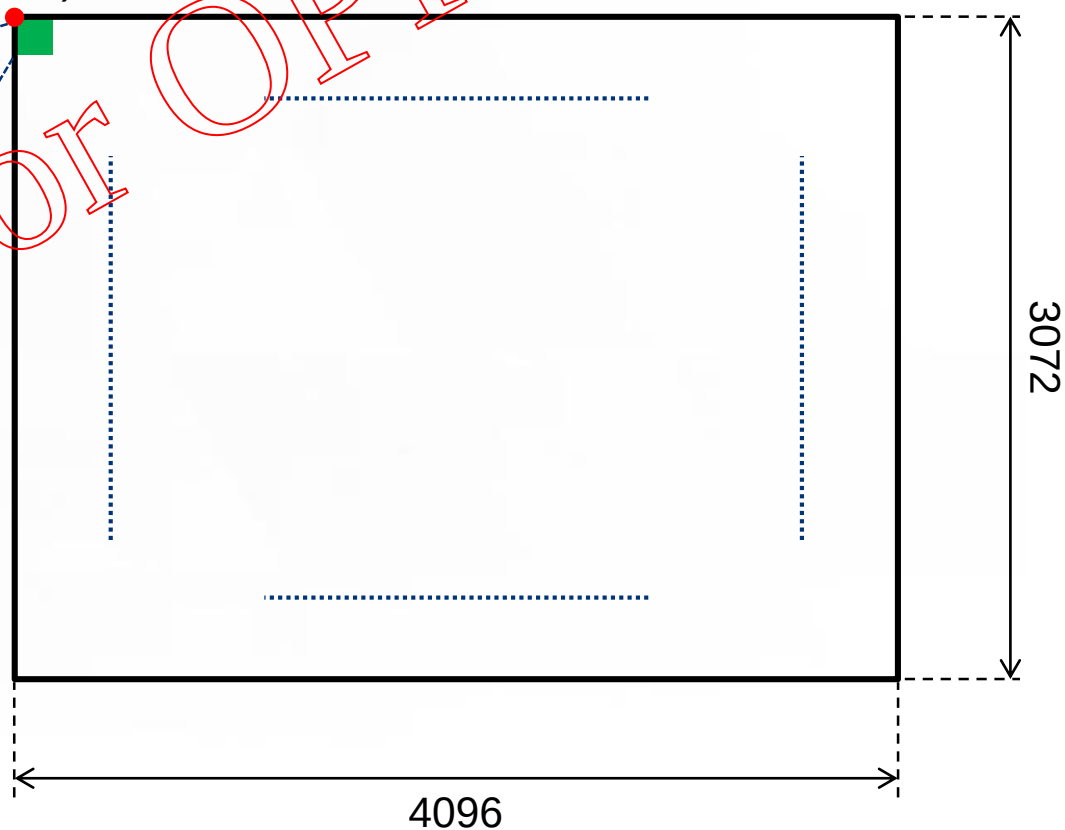
- ✓ QPD becomes DPD after vertical analog 2x binning.
- ✓ Binning size has two types of PD pixels, left and right PD pixel. Sensor has different options to output different amount of PD.

Normal

One block = multiple of 2x2 pixels
PD Density = 100%

	0	1	2	3	4	5	6	7
0	Blue	Green	Blue	Green	Blue	Green	Blue	Green
1	Green	Red	Green	Red	Green	Red	Green	Red
2	Blue	Green	Blue	Green	Blue	Green	Blue	Green
3	Green	Red	Green	Red	Green	Red	Green	Red
4	Blue	Green	Blue	Green	Blue	Green	Blue	Green
5	Green	Red	Green	Red	Green	Red	Green	Red
6	Blue	Green	Blue	Green	Blue	Green	Blue	Green
7	Green	Red	Green	Red	Green	Red	Green	Red

(0, 0)



PD Pattern and Alignment -- 4C2 Binning Size

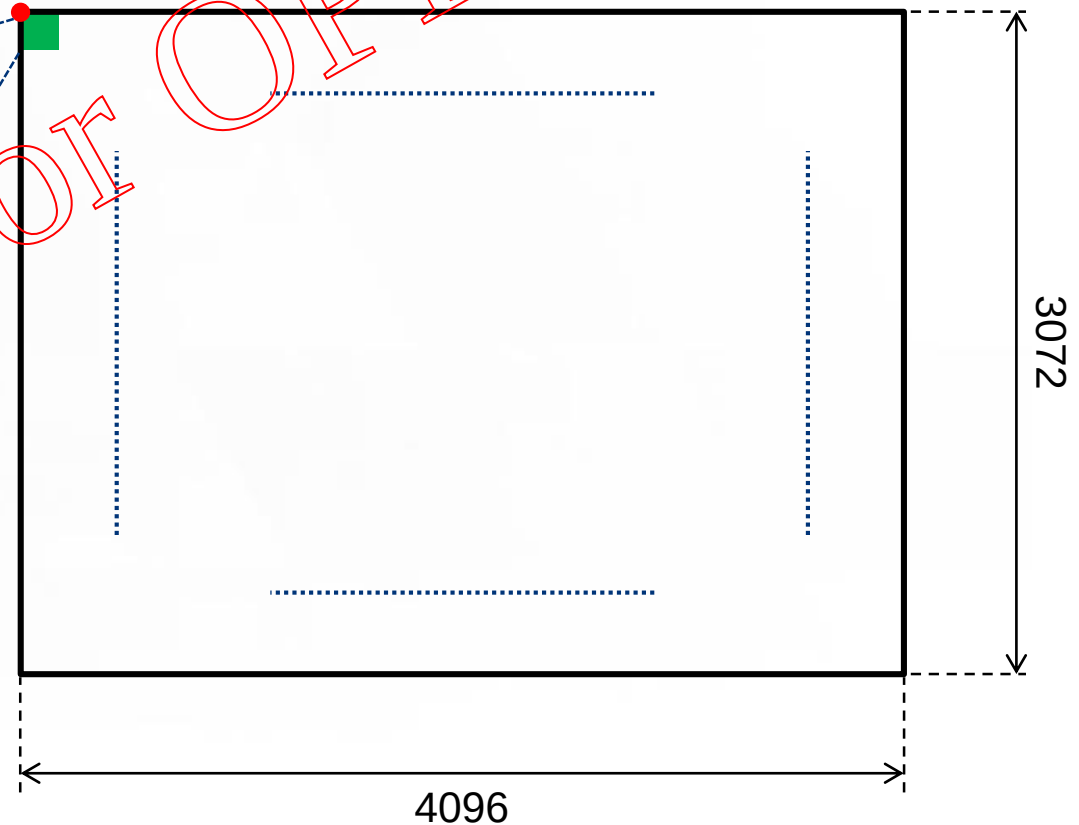
- ✓ QPD becomes DPD after vertical analog 2x binning.
- ✓ Binning size has two types of PD pixels, left and right PD pixel. Sensor has different options to output different amount of PD.

One block = multiple of 2x2 pixels
PD Density = 100%

Mirror&Flip

	0	1	2	3	4	5	6	7
0	Red	Green	Red	Green	Red	Green	Red	Green
1	Green	Blue	Green	Blue	Green	Blue	Green	Blue
2	Red	Green	Red	Green	Red	Green	Red	Green
3	Green	Blue	Green	Blue	Green	Blue	Green	Blue
4	Red	Green	Red	Green	Red	Green	Red	Green
5	Green	Blue	Green	Blue	Green	Blue	Green	Blue
6	Red	Green	Red	Green	Red	Green	Red	Green
7	Green	Blue	Green	Blue	Green	Blue	Green	Blue

(0, 0)



PD Pattern and Alignment -- 4x4 Binning Size

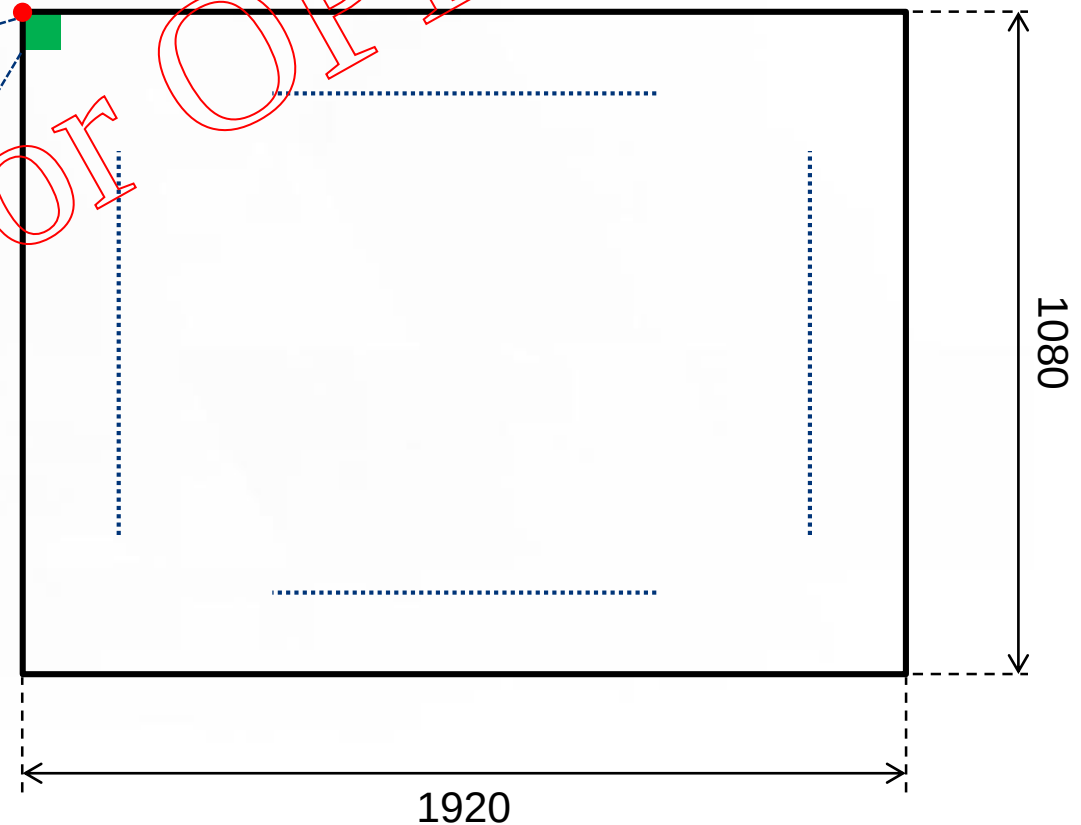
- ✓ QPD becomes DPD after vertical analog 4x binning.
- ✓ Binning size has two types of PD pixels, left and right PD pixel. Sensor has different options to output different amount of PD.

Normal

One block = multiple of 2x2 pixels
PD Density = 100%

	0	1	2	3	4	5	6	7
0	Blue	Green	Blue	Green	Blue	Green	Blue	Green
1	Green	Red	Green	Red	Green	Red	Green	Red
2	Blue	Green	Blue	Green	Blue	Green	Blue	Green
3	Green	Red	Green	Red	Green	Red	Green	Red
4	Blue	Green	Blue	Green	Blue	Green	Blue	Green
5	Green	Red	Green	Red	Green	Red	Green	Red
6	Blue	Green	Blue	Green	Blue	Green	Blue	Green
7	Green	Red	Green	Red	Green	Red	Green	Red

(0, 0)

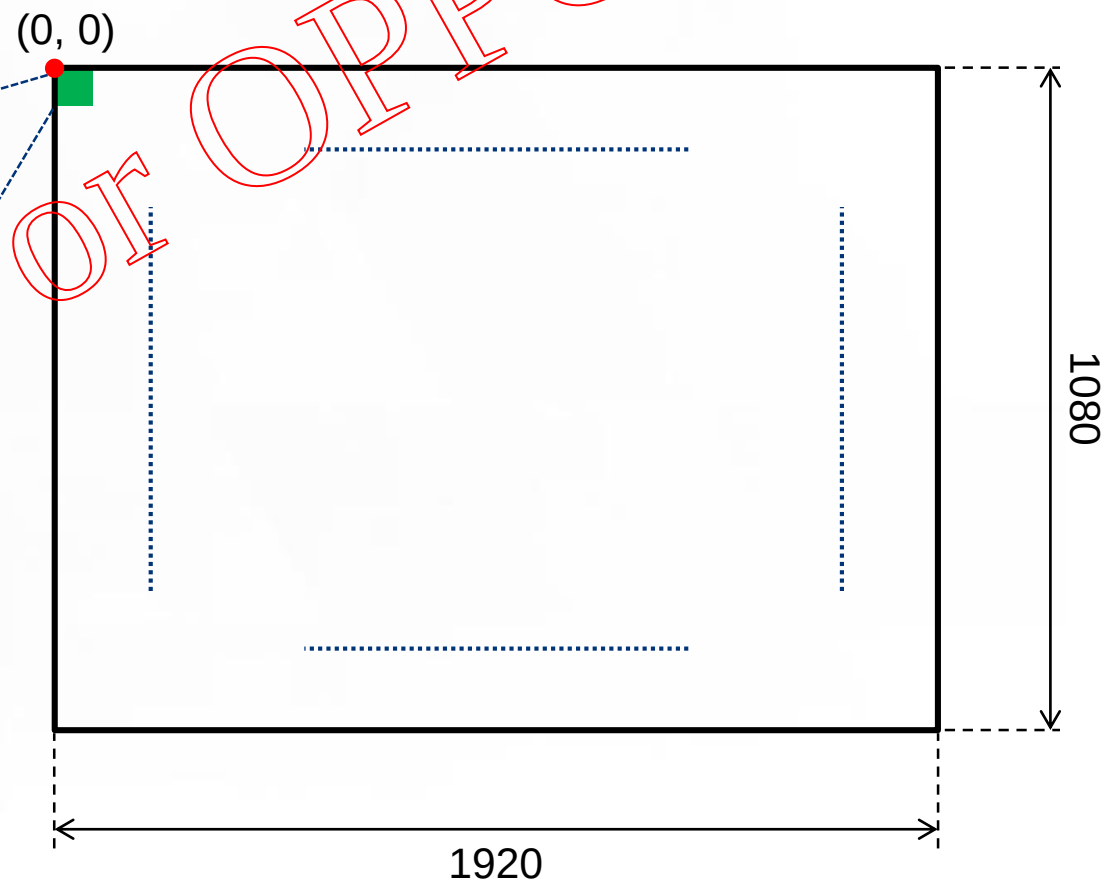


PD Pattern and Alignment -- 4x4 Binning Size

- ✓ QPD becomes DPD after vertical analog 4x binning.
- ✓ Binning size has two types of PD pixels, left and right PD pixel. Sensor has different options to output different amount of PD.

Mirror&Flip One block = multiple of 2x2 pixels
PD Density = 100%

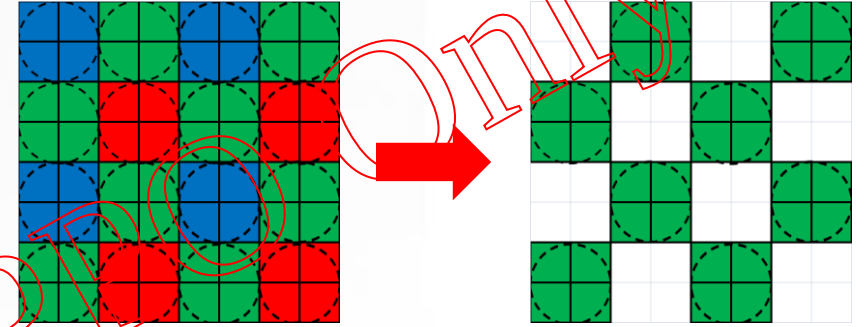
	0	1	2	3	4	5	6	7
0	Red	Green	Red	Green	Red	Green	Red	Green
1	Green	Blue	Green	Blue	Green	Blue	Green	Blue
2	Red	Green	Red	Green	Red	Green	Red	Green
3	Green	Blue	Green	Blue	Green	Blue	Green	Blue
4	Red	Green	Red	Green	Red	Green	Red	Green
5	Green	Blue	Green	Blue	Green	Blue	Green	Blue
6	Red	Green	Red	Green	Red	Green	Red	Green
7	Green	Blue	Green	Blue	Green	Blue	Green	Blue



PD Data -- Full Size (50MP) 8192x6144

G output

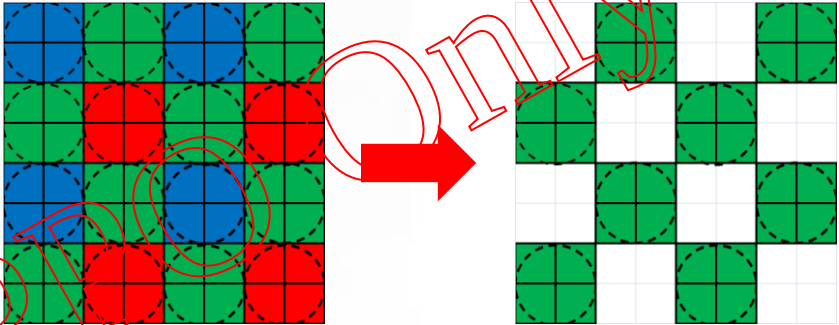
- ✓ Sensor does full or cropping in analog domain. QPD re-mosaic for image converts QPD to Bayer pattern. Sensor outputs 2x2 PD original G data through MIPI DT/VC.
- ✓ PD data sequence is same as physical pixel sequence like one row G1, G2, G1, G2... then next row is G3, G4, G3, G4...
- ✓ PD data amount options
 - Windowing
 - VSkip2
 - HSkip2



G1	G2
G3	G4

PD Data -- Full Size (50MP) 8192x6144

G output



0x45c0[2] = 0 indicates G data output

Image Size	H Skip2 0x45c1[0]	V Skip2 0x45c1[2]	V Skip4 0x45c1[3]	PD Size
8192x6144				4096x6144
		✓		4096x3072
			✓	4096x1536
	✓			2048x6144
	✓	✓		2048x3072
	✓		✓	2048x1536

G1	G2
G3	G4

PD Data -- Full Size (50MP) 8192x6144

Y or Combined G output

- Sensor can use B/Gb/Gr/R to calculate Y or combined G through programmable coefficients as below.

Programmable coefficients Cb (0x45cd), Cgb (0x45ce), Cgr (0x45cf), Cr (0x45d0).

Total C = Cb + Cgb + Cgr + Cr. Total C must be 0x40.

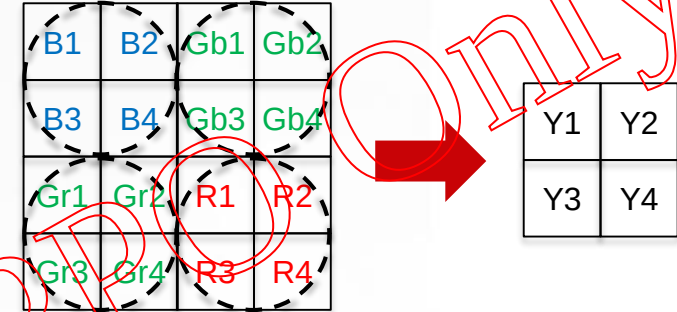
$$Y1 = (B1 * Cb + Gb1 * Cgb + Gr1 * Cgr + R1 * Cr) / C$$

$$Y2 = (B2 * Cb + Gb2 * Cgb + Gr2 * Cgr + R2 * Cr) / C$$

$$Y3 = (B3 * Cb + Gb3 * Cgb + Gr3 * Cgr + R3 * Cr) / C$$

$$Y4 = (B4 * Cb + Gb4 * Cgb + Gr4 * Cgr + R4 * Cr) / C$$

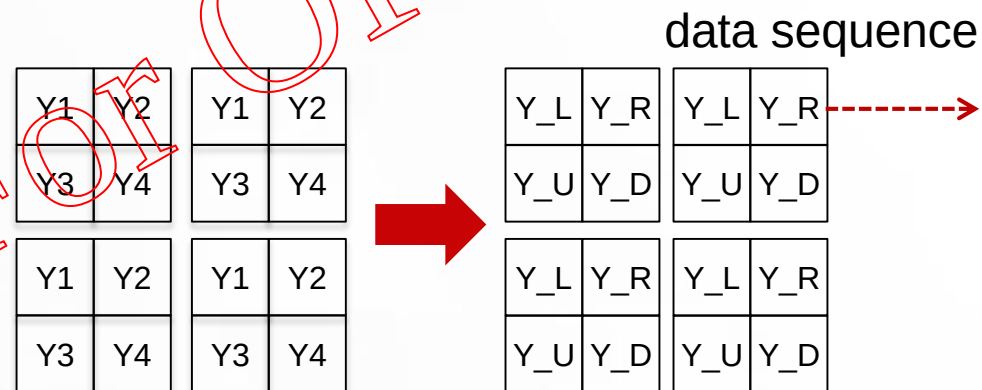
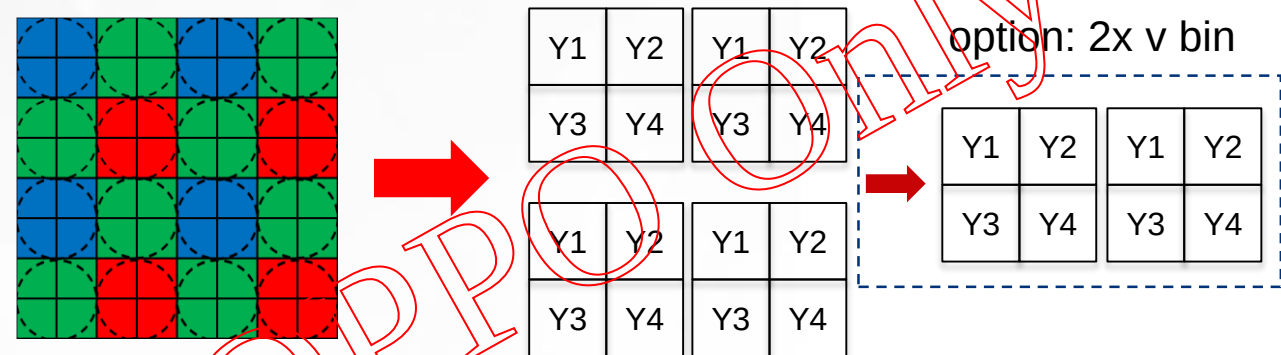
when Cb, Cr is 0 and Cgb=Cgr=0x20, the data is combined G.



PD Data -- Full Size (50MP) 8192x6144

Y or Combined G output

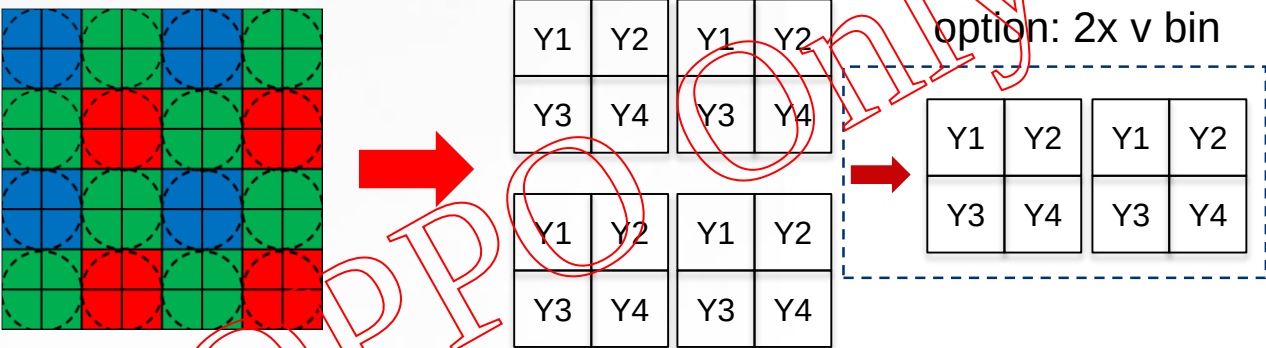
- ✓ PD data sequence is same as physical pixel sequence like one row Y1, Y2, Y1, Y2... then next row is Y3, Y4, Y3, Y4...
- ✓ PD data amount options
 - Windowing
 - VBin2
 - VSkip2, VSkip4
 - HSkip2
 - Convert Y1/Y2/Y3/Y4 to Y_L/Y_R/Y_U/Y_D. PD data sequence is L/R line first then U/D line. L/R and U/D lines have different VC or DT. If do VSkip then sensor can only output L/R (or U/D) data.
0x45c3[0] is conversion enable.
0x45c3[1]: 0 is sum, 1 is average in conversion.



PD	Average	Sum
Y_L	$(Y1 + Y3) / 2$	$Y1 + Y3$
Y_R	$(Y2 + Y4) / 2$	$Y2 + Y4$
Y_U	$(Y1 + Y2) / 2$	$Y1 + Y2$
Y_D	$(Y3 + Y4) / 2$	$Y3 + Y4$

PD Data -- Full Size (50MP) 8192x6144

Y or Combined G output

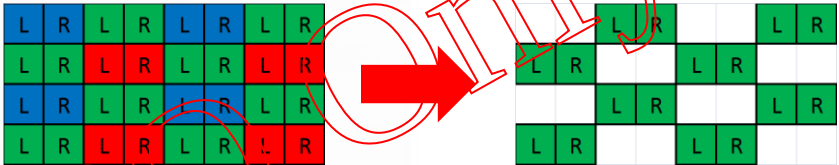


0x45c0[2] = 1 indicates Y or Combined G output
0x45c2[3] = 1 is average in V Bin2, 0 is sum

Image Size	H Skip2 0x45c1[0]	V Bin2 0x45c0[3]	V Skip2 0x45c1[2]	V Skip4 0x45c1[3]	PD Size
8192x6144					4096x3072
		✓	or (✓)		4096x1536
		✓	✓	or (✓)	4096x768
	✓				2048x3072
	✓	✓			2048x1536
	✓	✓	✓		2048x768
	✓	✓		✓	2048x384

PD Data -- 4C2 Binning Size (12.5MP) 4096x3072

G output



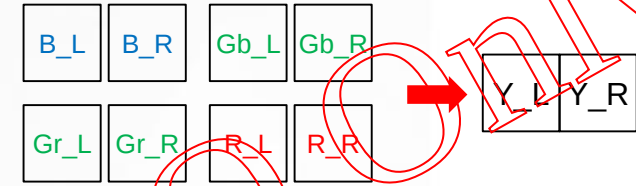
- Sensor does vertical binning in analog domain. Then PD only has L/R phase information. Sensor outputs 2x1 PD original G data through MIPI DT/VC.
 - PD data amount options
- 0x45c0[2] = 0 indicates G data output

- Windowing
- VSkip2
- HSkip2

Image Size	H Skip2 0x45c1[0]	V Skip2 0x45c1[2]	V Skip4 0x45c1[3]	PD Size
4096x3072				4096x3072
		√		4096x1536
			√	4096x768
	√			2048x3072
	√	√		2048x1536
	√		√	2048x768

PD Data -- 4C2 Binning Size (12.5MP) 4096x3072

Y or Combined G output



- ✓ Sensor can use B/Gb/Gr/R to calculate Y or combined G through programmable coefficients as below.

Programmable coefficients Cb (0x45cd), Cgb (0x45ce), Cgr (0x45cf), Cr (0x45d0).

Total C = Cb + Cgb + Cgr + Cr. Total C must be 0x40.

$$Y_L = (B_L * Cb + Gb_L * Cgb + Gr_L * Cgr + R_L * Cr) / C$$

$$Y_R = (B_R * Cb + Gb_R * Cgb + Gr_R * Cgr + R_R * Cr) / C$$

when Cb, Cr is 0 and Cgb=Cgr=0x20, the data is combined G.

- ✓ PD data amount options

- Windowing
- VBin2
- VSkip2, VSkip4
- HSkip2

PD Data -- 4C2 Binning Size (12.5MP) 4096x3072

Y or Combined G output

0x45c0[2] = 1 indicates Y or Combined G output

0x45c2[3] = 1 is average in V Bin2, 0 is sum

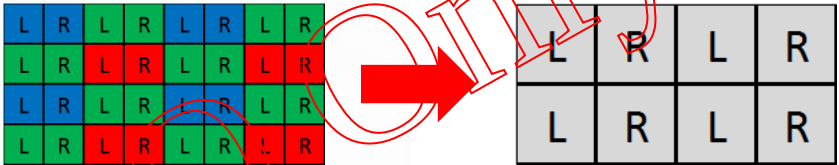
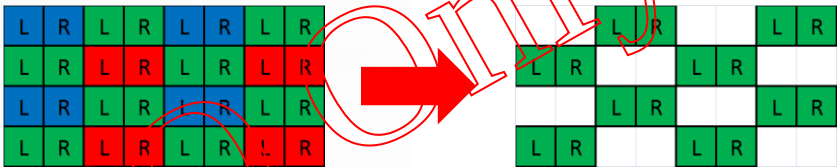


Image Size	H Skip2 0x45c1[0]	V Bin2 0x45c0[3]	V Skip2 0x45c1[2]	V Skip4 0x45c1[3]	PD Size
4096x3072					4096x1536
		✓	or (✓)		4096x768
		✓	✓	or (✓)	4096x384
	✓				2048x1536
	✓	✓			2048x768
	✓	✓	✓		2048x384
	✓	✓		✓	2048x192

PD Data -- 16C2 Binning Size (2MP) 1920x1080

G output



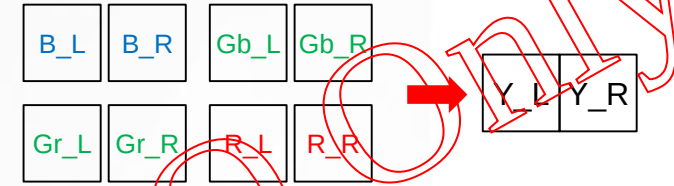
- Sensor does vertical binning in analog domain. Then PD only has L/R phase information. Sensor outputs 2x1 PD original G data through MIPI DT/VC.
 - PD data amount options
- 0x45c0[2] = 0 indicates G data output

- Windowing
- VSkip2
- HSkip2

Image Size	H Skip2 0x45c1[0]	V Skip2 0x45c1[2]	V Skip4 0x45c1[3]	PD Size
1920x1080				3840x1080
		√		3840x540
			√	3840x270
	√			1920x1080
	√	√		1920x540
	√		√	1920x270

PD Data -- 16C2 Binning Size (2MP) 1920x1080

Y or Combined G output



- Sensor can use B/Gb/Gr/R to calculate Y or combined G through programmable coefficients as below. Programmable coefficients Cb (0x45cd), Cgb (0x45ce), Cgr (0x45cf), Cr (0x45d0).

Total C = Cb + Cgb + Cgr + Cr. Total C must be 0x40.

$$Y_L = (B_L * Cb + Gb_L * Cgb + Gr_L * Cgr + R_L * Cr) / C$$

$$Y_R = (B_R * Cb + Gb_R * Cgb + Gr_R * Cgr + R_R * Cr) / C$$

when Cb, Cr is 0 and Cgb=Cgr=0x20, the data is combined G.

- PD data amount options

- Windowing
- VBin2
- VSkip2, VSkip4
- HSkip2

PD Data -- 16C2 Binning Size (2MP) 1920x1080

Y or Combined G output

0x45c0[2] = 1 indicates Y or Combined G output

0x45c2[3] = 1 is average in V Bin2, 0 is sum

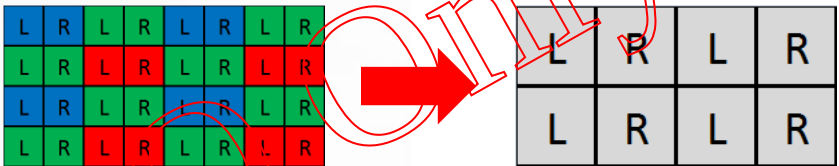
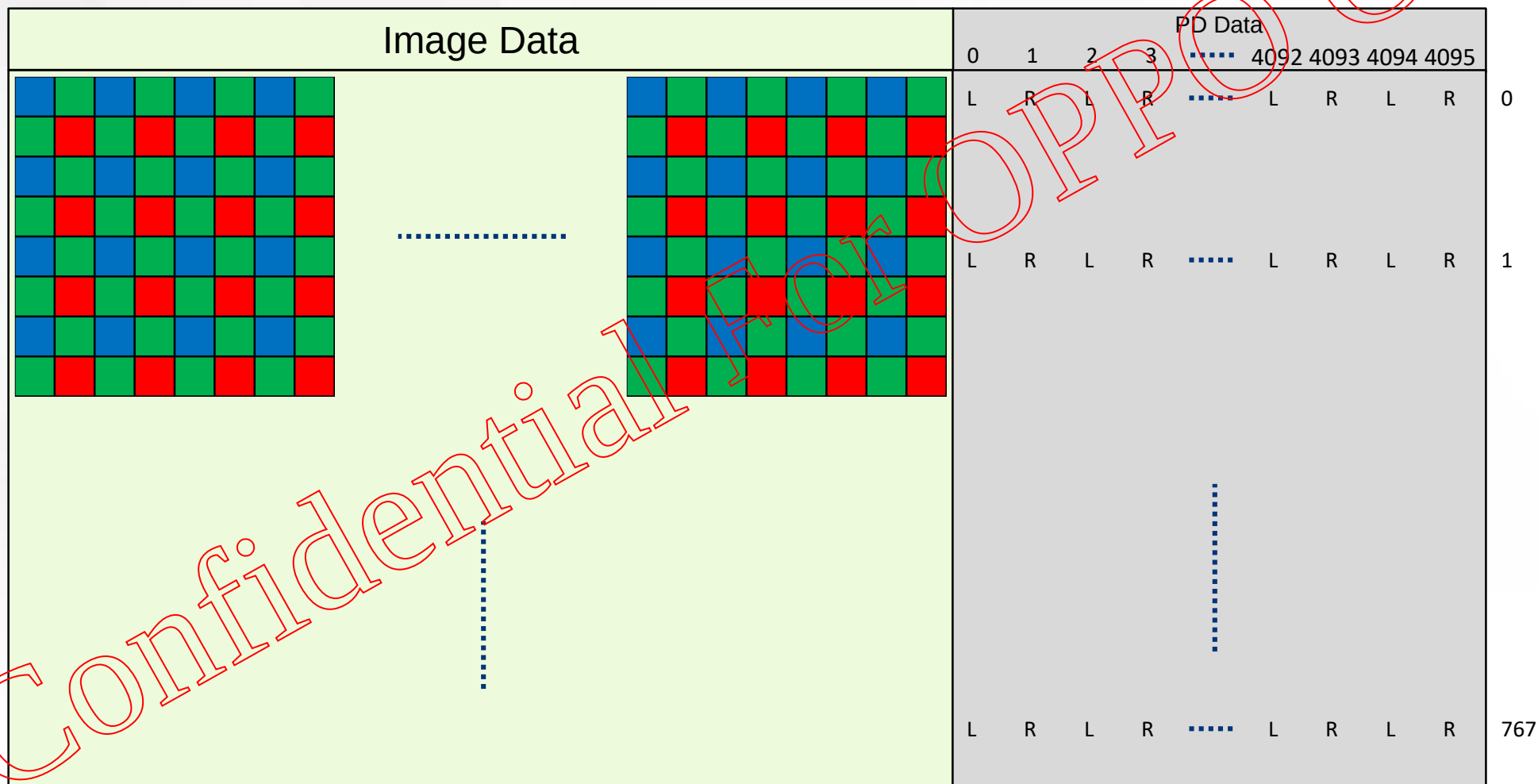


Image Size	H Skip2 0x45c1[0]	V Bin2 0x45c0[3]	V Skip2 0x45c1[2]	V Skip4 0x45c1[3]	PD Size
1920x1080					3840x540
		✓	or (✓)		3840x270
		✓	✓	or (✓)	3840x135
	✓				1920x540
	✓	✓			1920x270
	✓	✓	✓		1920x135
	✓			✓	1920x135

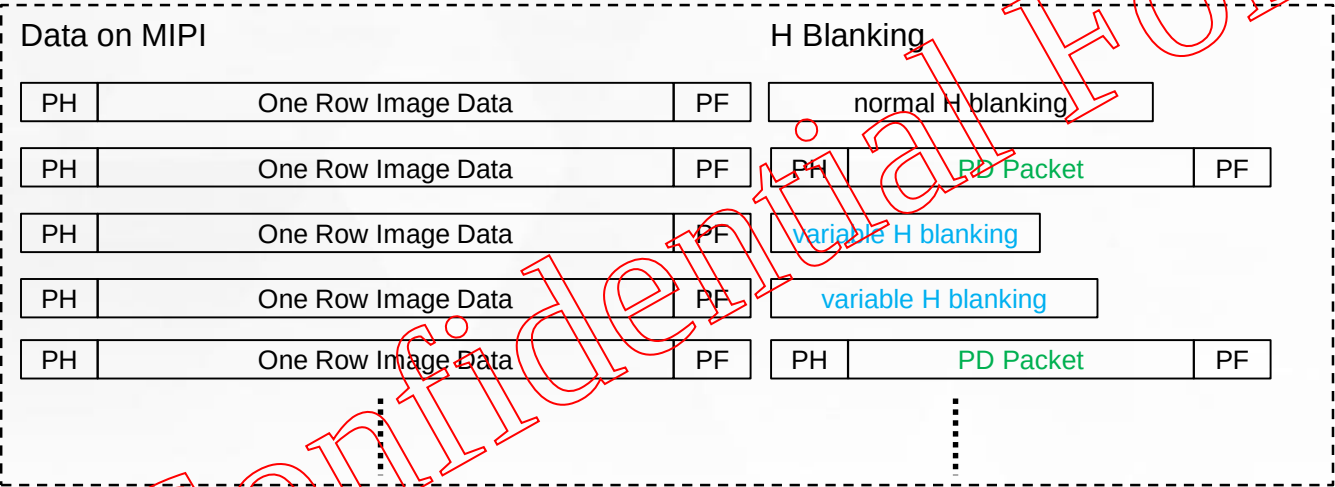
PD Data Sequence -- 4C2 Binning Size (12.5MP) 4096x3072

- ✓ PD data is sent in h blanking after one line image data. The PD amount (PD packet size) in one h blanking is programmable. Below diagram shows maximum PD packet size. Total PD amount is 4096x768.



PD Data Timing

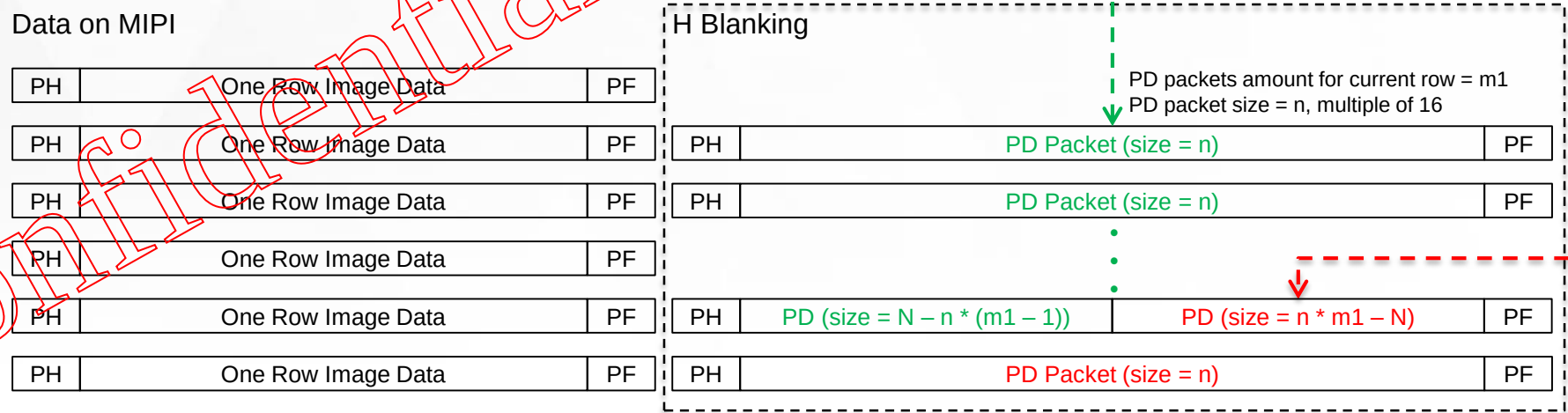
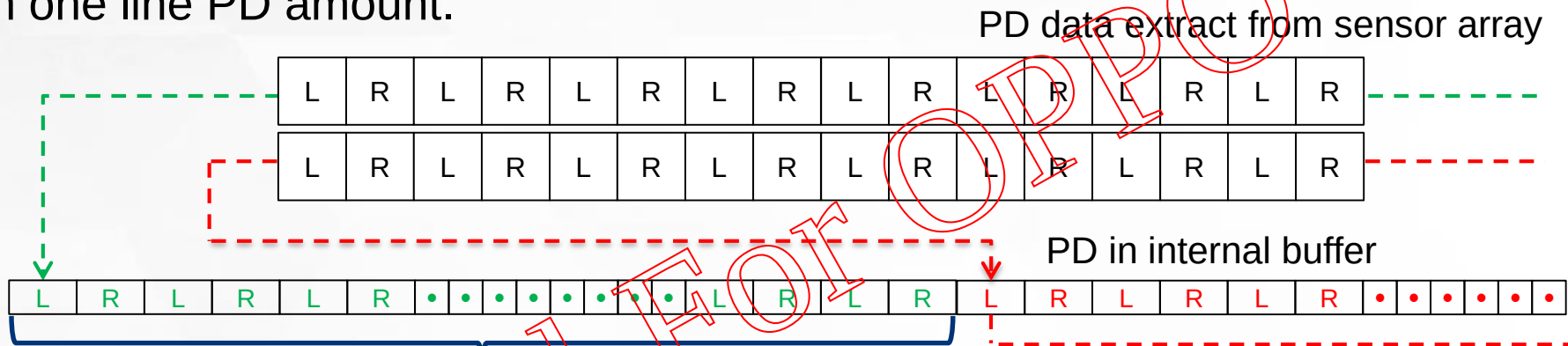
- Depends on h blanking time and MIPI speed, different data packet size affects normal image lines h blanking time. When PD packet can't be done in normal h blanking, it takes more time to transfer PD. Then h blanking of following images lines becomes short. As below diagram shown after PD packet done, the following images lines have variable h blanking.



PD Packet Data

- No padding in PD packet data. So if one line PD amount isn't multiple of PD packet size, different lines' PD are merged in one PD packet. Platform may not support it. So please change PD packet size based on one line PD amount.

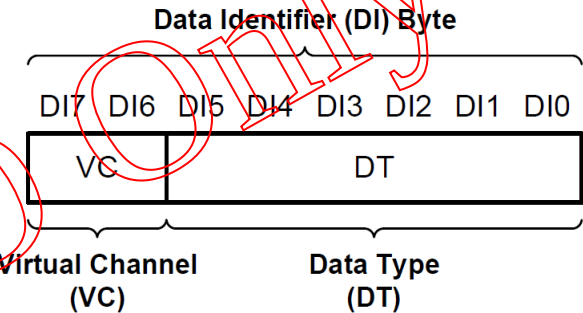
{0x4640, 0x4641}
is PD packet size
registers. The
packet size is
register value * 16.
Unit is pixel.



PD Packet

Register 0x4684[5:0] is PD data type. In current setting it is 0x2b for VC mode and 0x19 for DT mode.

Register 0x4680[6:4], 0x4813, 0x4836 is to decide PD VC value.



* Refer to the MIPI alliance CSI-2 V2.1

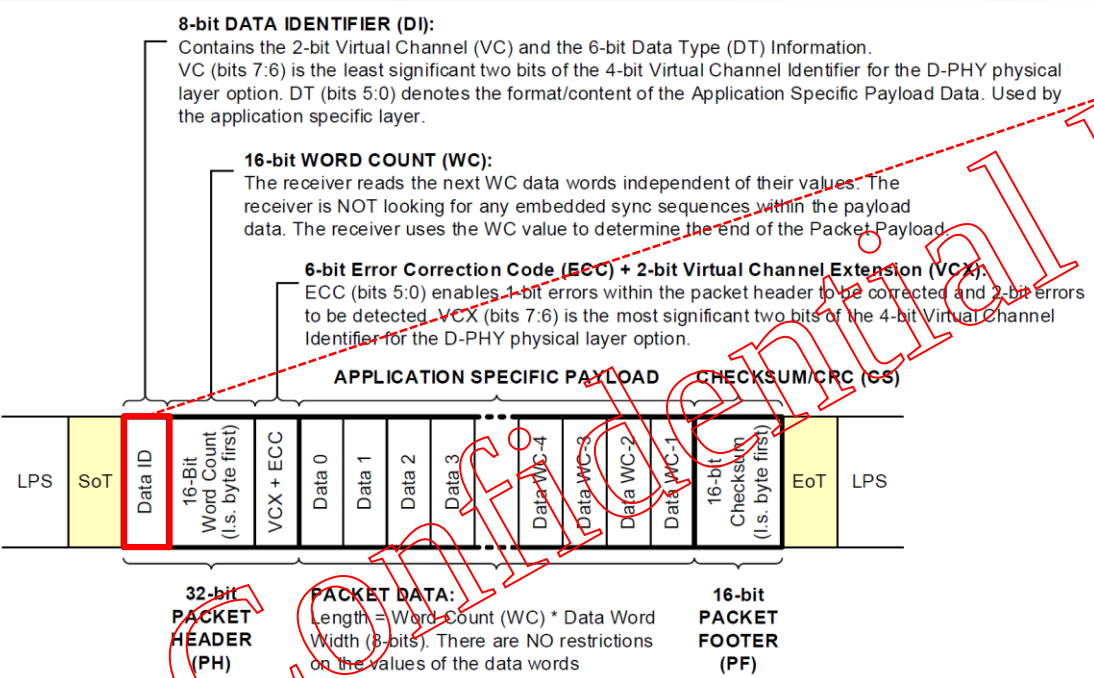


Figure 52 Long Packet Structure for D-PHY Physical Layer Option

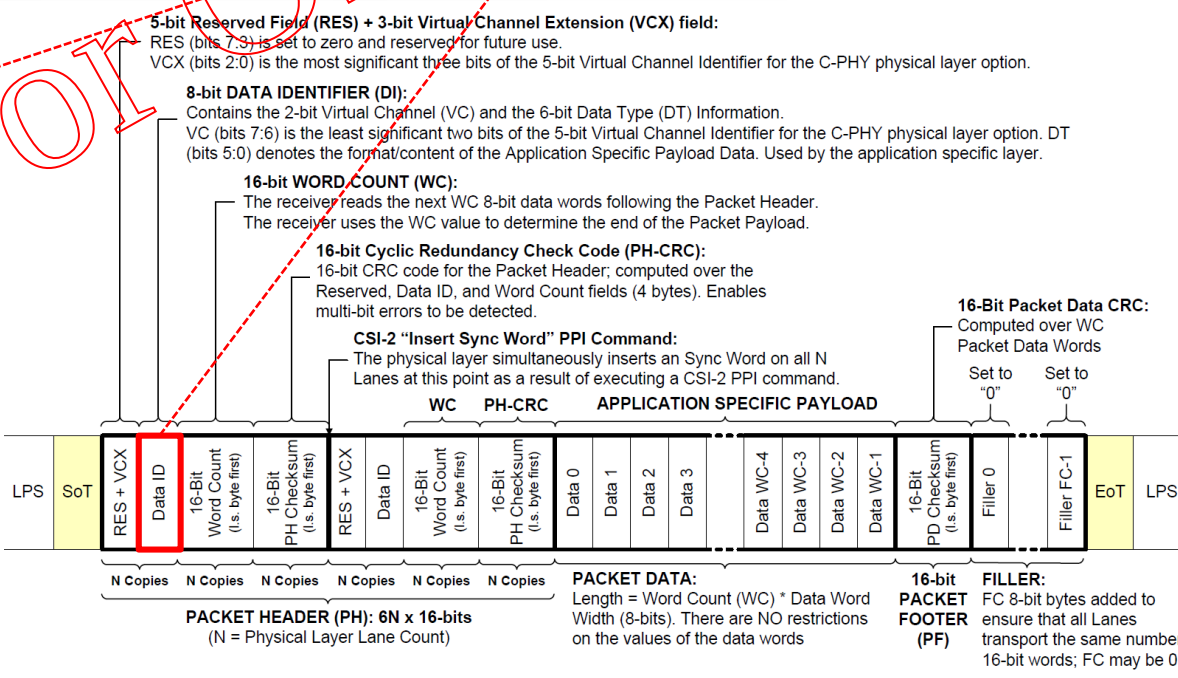


Figure 53 Long Packet Structure for C-PHY Physical Layer Option

Thank You

Confidential For OPPO Only